

NURS 1016 Anatomy & Physiology I

Chapter 1: An Introduction to the Human Body

➤ Today's Lesson

- Definition of **Anatomy** and **Physiology**
- Levels of structural organization
- Necessary life functions & survival needs
- Organ systems (briefly)
- Homeostasis (important!)
- Language of anatomy (anatomical position, directional /regional terms, body planes)

➤ Overview of A&P

- **Anatomy**
 - Study of the structure of the human body
- Subdivisions:
 - i. Gross (e.g., regional, systemic, and surface anatomy)
 - ii. Microscopic (e.g., histology)

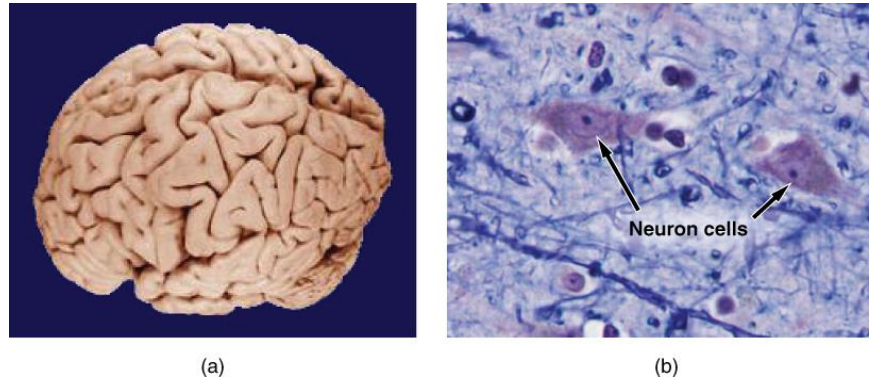


Fig. 1.2 Gross and Microscopic Anatomy

- (a) Gross anatomy considers large structures such as the brain.
- (b) Microscopic anatomy can deal with the same structures, though at a different scale. This is a micrograph of nerve cells from the brain. LM $\times 1600$. (credit a: "WriterHound"/Wikimedia Commons; credit b: Micrograph provided by the Regents of University of Michigan Medical School © 2012)

➤ Overview of A&P

- **Physiology**
 - Study of the function of the human body
 - Subdivisions based on organ systems (e.g., renal or cardiovascular physiology)
 - cellular and biochemical pathways

➤ Principle of Complementarity

- Anatomy and physiology are inseparable
 - Function always reflects structure
 - Physiology is dependent on the anatomy

➤ Structural Organization

- Biochemical (Chapter 2)
- Cellular (Chapter 3)
- Tissue (groups of similar cells) (Chapter 4)
- Organ (one or more types of tissues)
- Organ System (organs that physiologically linked together)
- Organismal (all the organ systems in the human body working together)

➤ Structural Organization

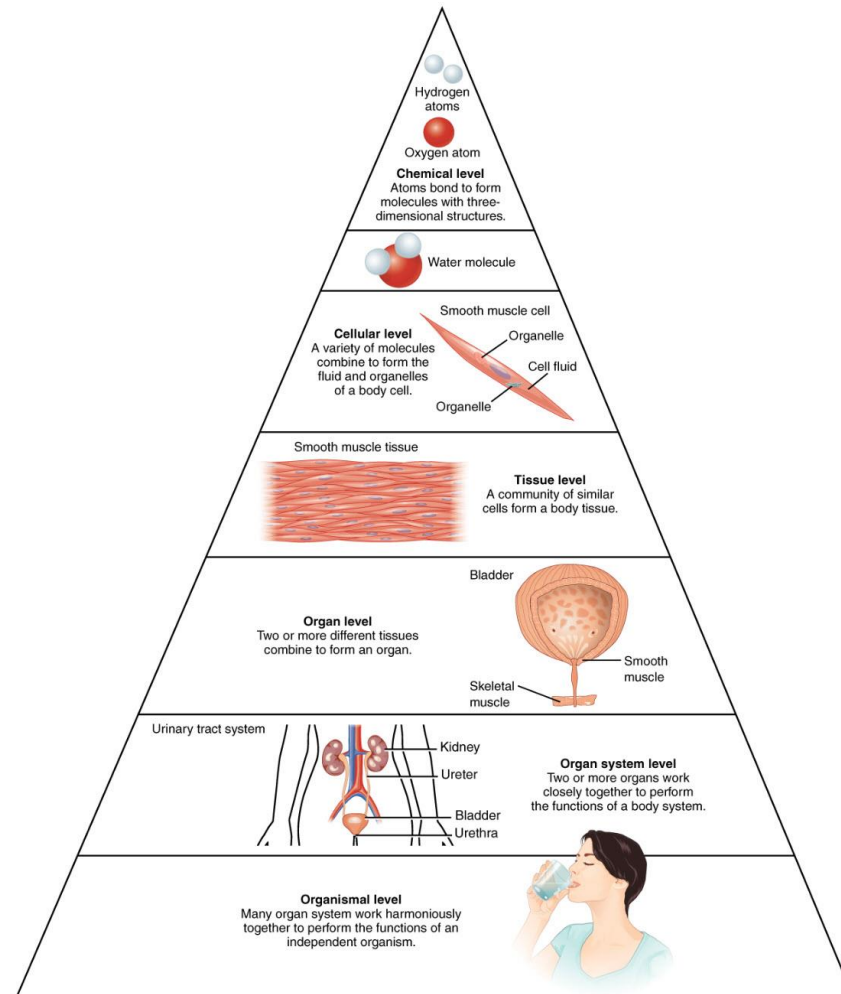


Fig. 1.3 Levels of Structural Organization of the Human Body

➤ Necessary Life Functions

1. Metabolism: anabolism and catabolism
2. Responsiveness
3. Movement
4. Development, growth and reproduction

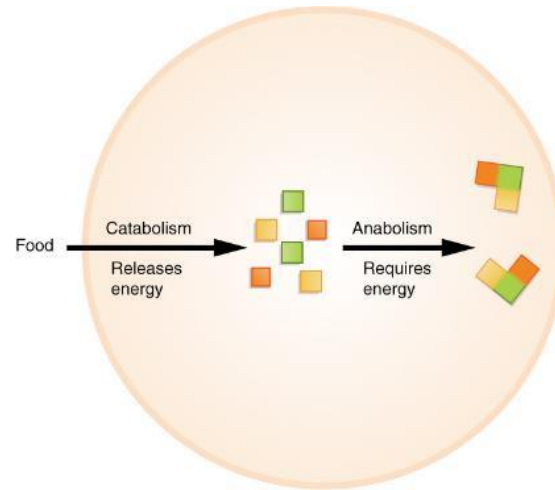


Fig. 1.6 Metabolism

➤ Organ Systems

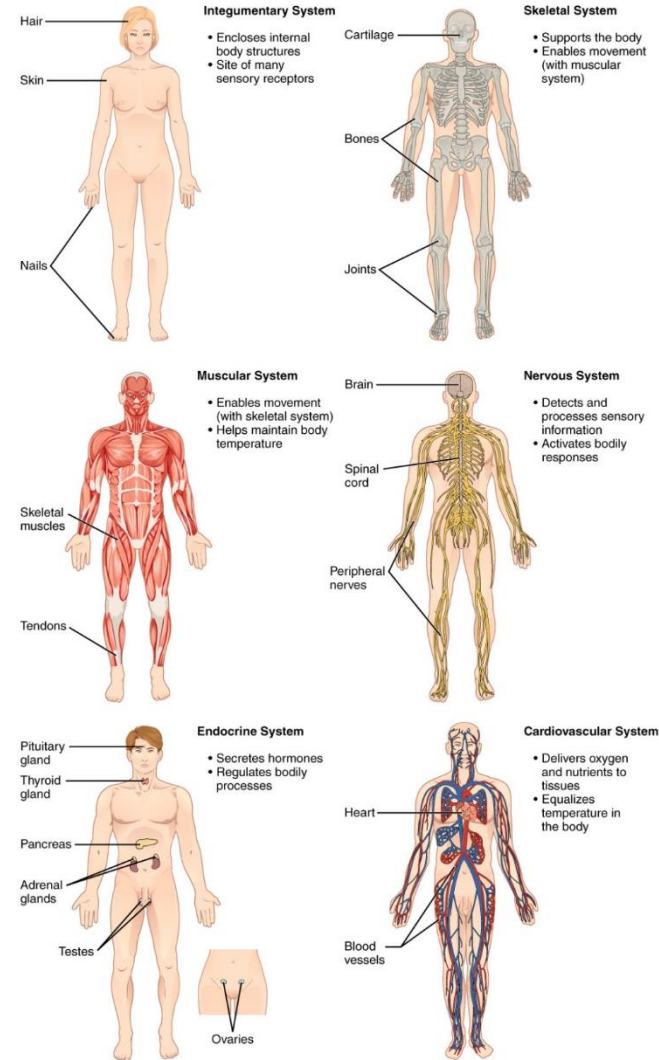


Fig. 1.4 Organ Systems of the Human Body

➤ Organ Systems

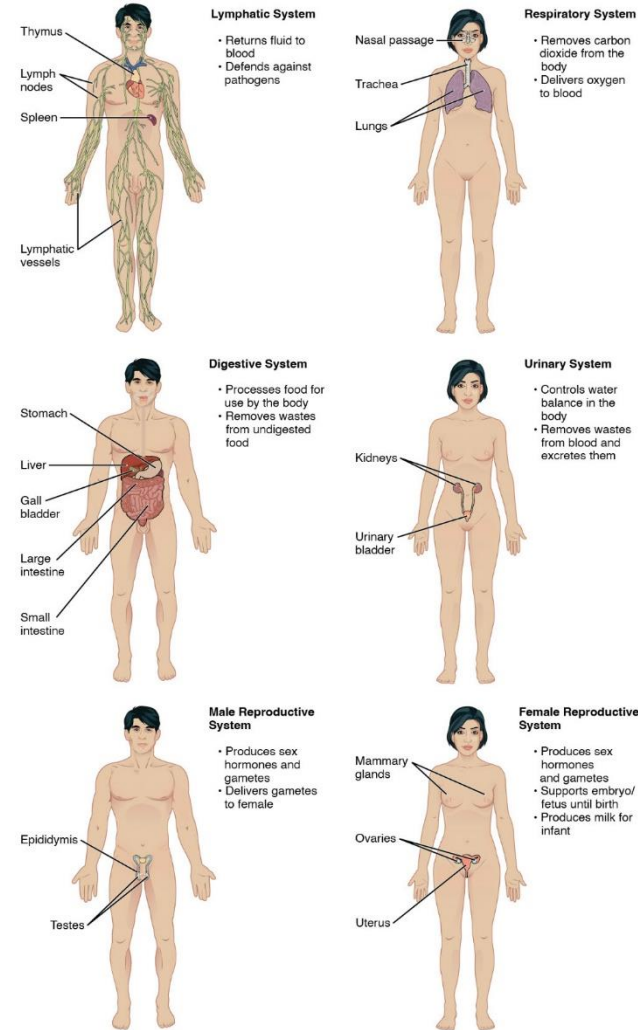


Fig. 1.4 Organ Systems of the Human Body (cont'd)

➤ Requirements for Human Life

- Oxygen
- Nutrients
- Narrow range of temperature

➤ Homeostasis

- Maintenance of a stable internal environment despite constantly changing conditions
- Dynamic equilibrium of certain physiological variables (e.g., BP, pH, temperature)
- Physiological variable has a set point or normal range
- Requires: receptor, control centre, and effector

➤ Components of Homeostatic System

1. Receptor

- monitors physiological variable (e.g., BP, pH, Temperature)
- sends info to control centre if the variable deviates from set point

2. Control center

- determines set point (or range) of variable
- receives input via afferent pathways from receptor
- sends instructions via efferent pathways to effector to reestablish homeostasis

3. Effector

- receives instructions from control center
- responds to reestablish homeostasis
- response is terminated by negative feedback

➤ Negative Feedback Loop

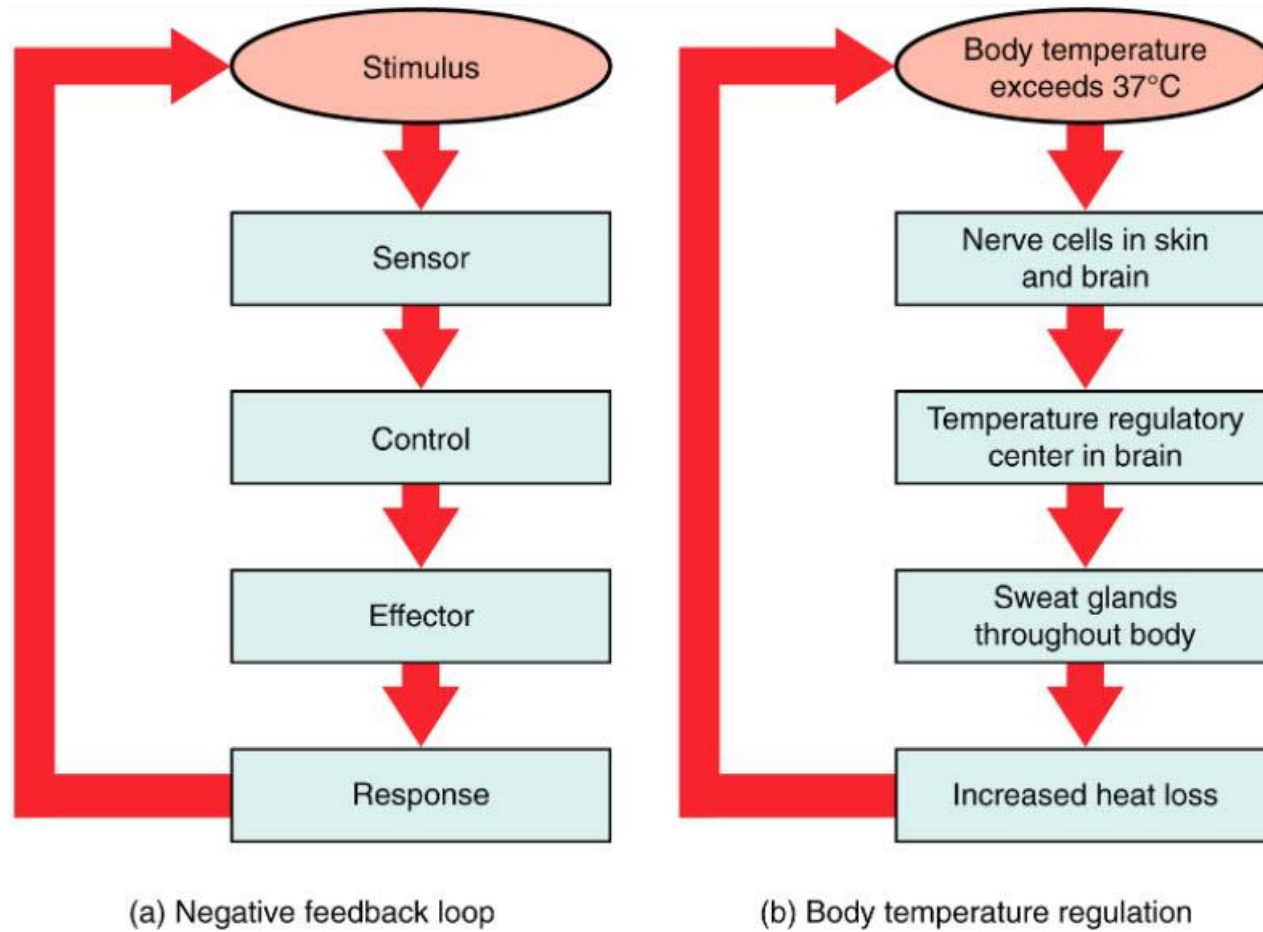
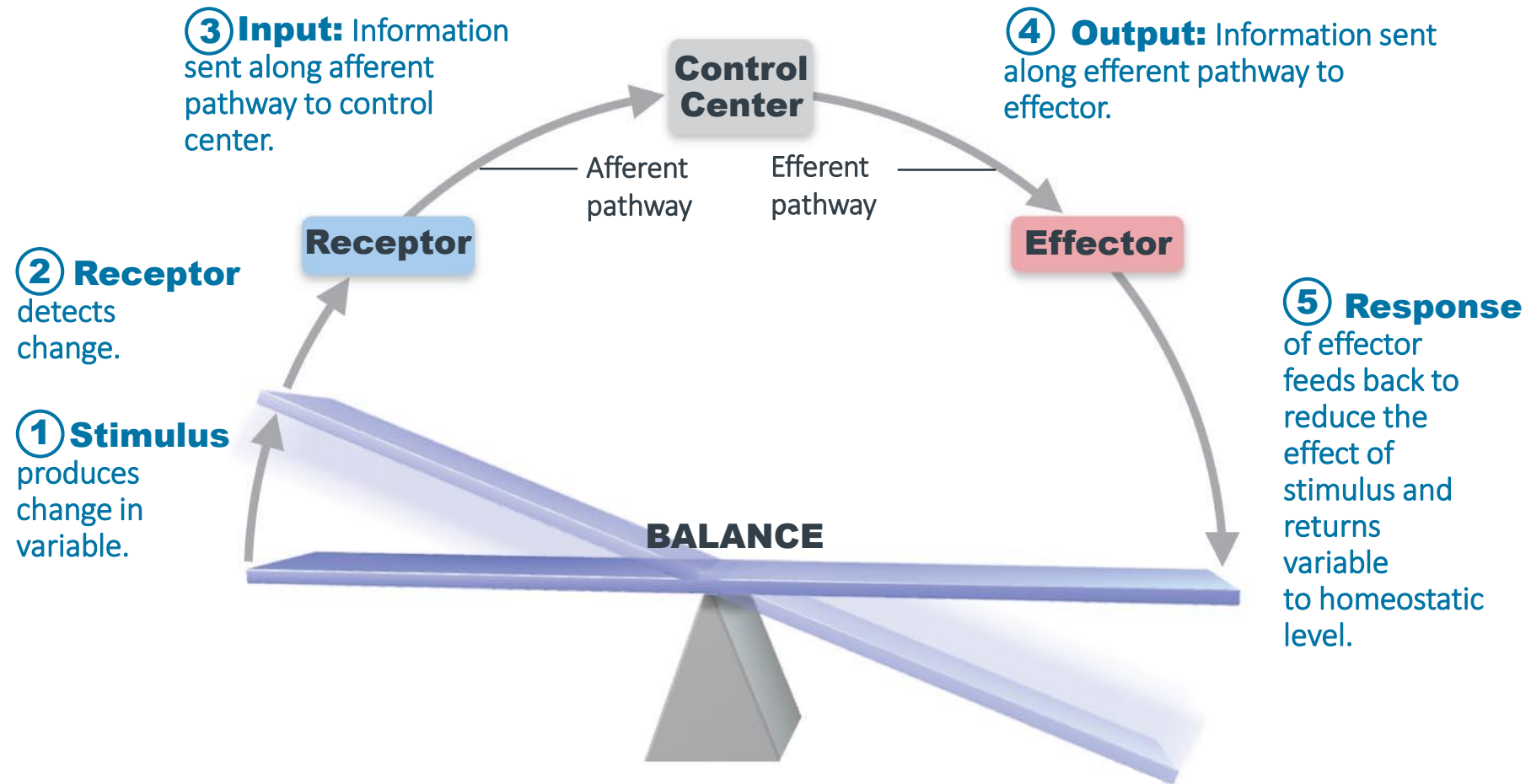


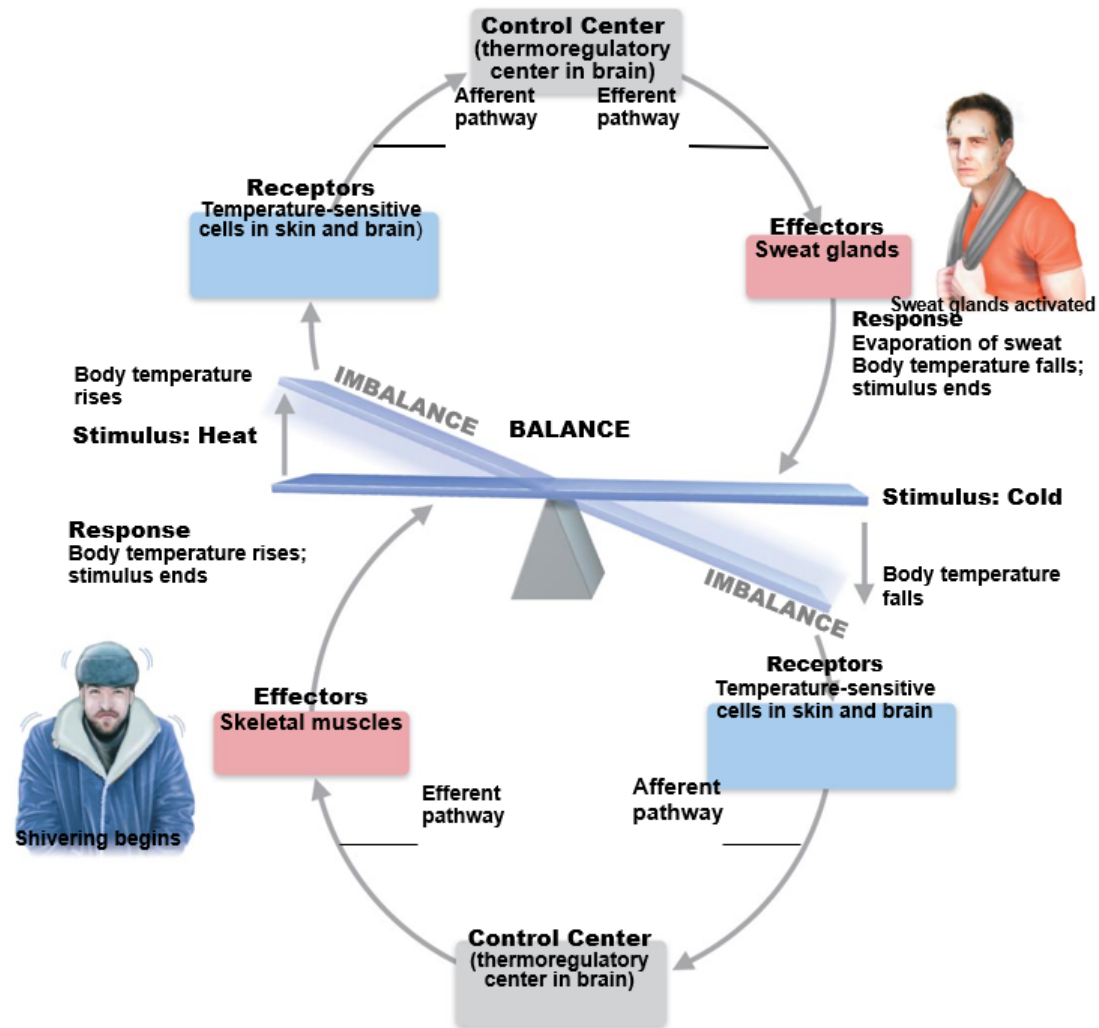
Figure 1.10 Negative Feedback System In a negative feedback system, a stimulus—a deviation from a set point—is resisted through a physiological process that returns the body to homeostasis. (a) A negative feedback system has five basic parts. (b) Body temperature is regulated by negative feedback.

➤ Homeostasis



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➤ Homeostasis



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➤ Positive Feedback

- Response of the effector amplifies the initial change in the variable
- Rare compared to negative feedback
- Examples: labour contractions, platelet plug formation

➤ Anatomical Position

- Standard anatomical body position
 - Body erect
 - Feet slightly apart
 - Palms facing forward
 - Thumbs point away from body
- Always use directional terms as if body is in anatomical position
- Right and left refer to body being viewed, not those of observer

➤ Regional Terms

- Two major divisions of body:
 - **Axial**
 - Head, neck, and trunk
 - **Appendicular**
 - Limbs
- Regional terms designate specific areas of the body

➤ Regions of the Human Body

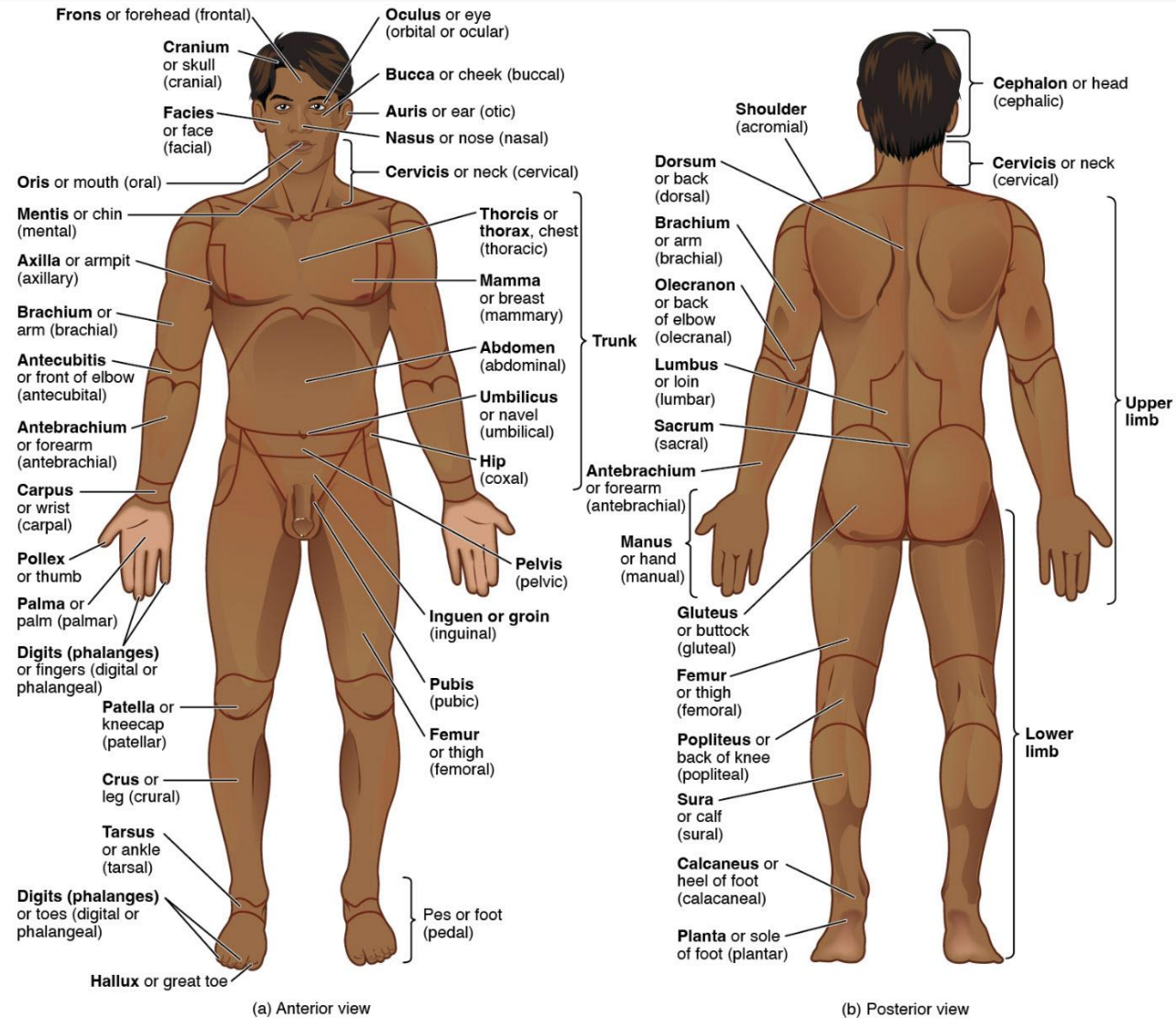


Figure 1.12 Regions of the Human Body The human body is shown in anatomical position in an (a) anterior view and a (b) posterior view. The regions of the body are labeled in boldface.

➤ Directional Terms

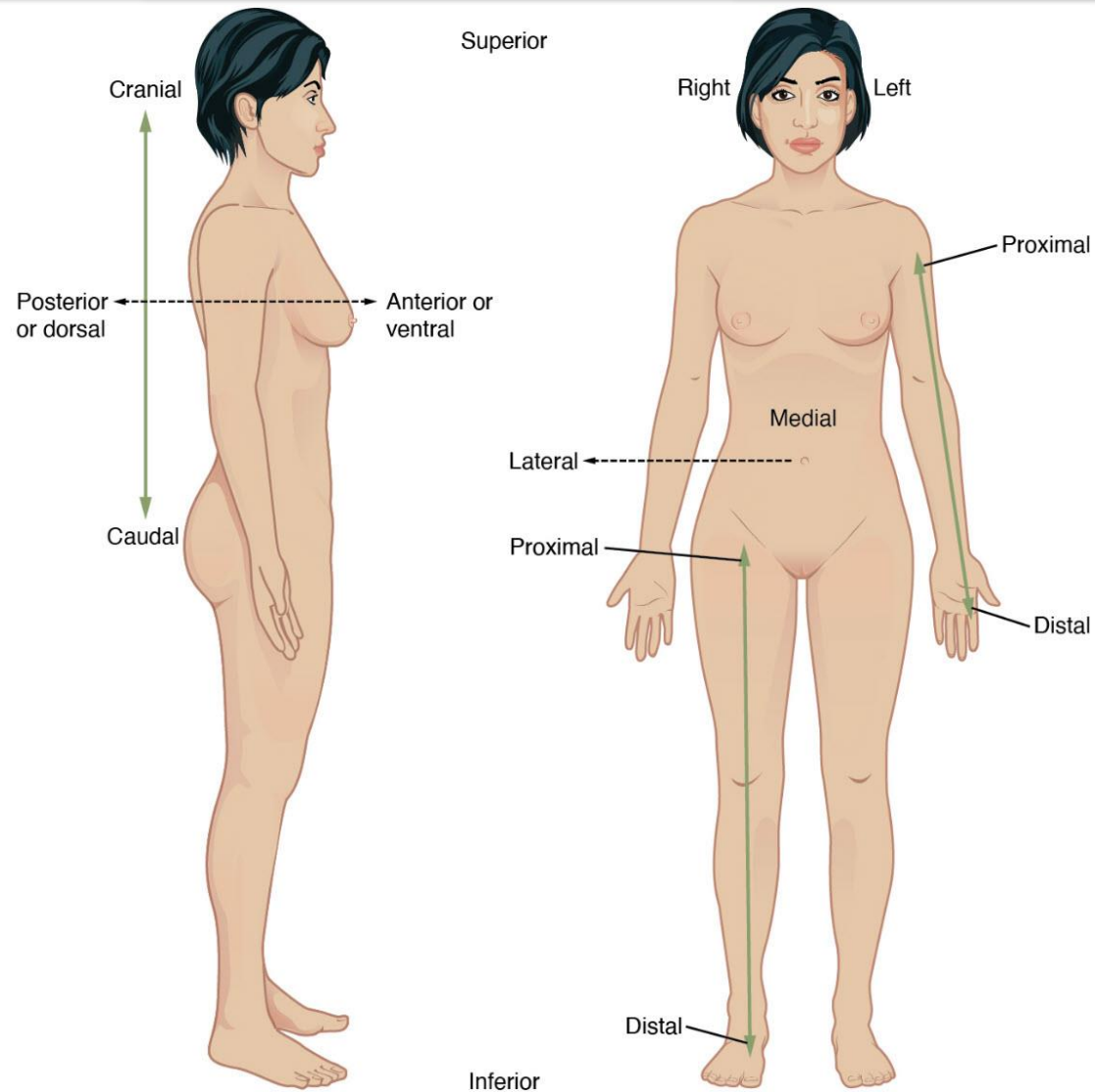


Fig. 1.13 Directional Terms Applied to the Human Body Paired directional terms are shown as applied to the human body.

➤ Directional Terms

- **Anterior** (or **ventral**) Describes the front or direction toward the front of the body.
- **Posterior** (or **dorsal**) Describes the back or direction toward the back of the body.
- **Superior** describes a position above or higher than another part of the body proper.
- **Inferior** describes a position below or lower than another part of the body proper; near or toward the tail (in humans, the coccyx, or lowest part of the spinal column).

➤ Directional Terms

- **Lateral** describes the side or direction toward the side of the body.
- **Medial** describes the middle or direction toward the middle of the body.
- **Proximal** describes a position in a limb that is nearer to the point of attachment or the trunk of the body.
- **Distal** describes a position in a limb that is farther from the point of attachment or the trunk of the body.
- **Superficial** describes a position closer to the surface of the body.
- **Deep** describes a position farther from the surface of the body.

➤ Planes of the Body

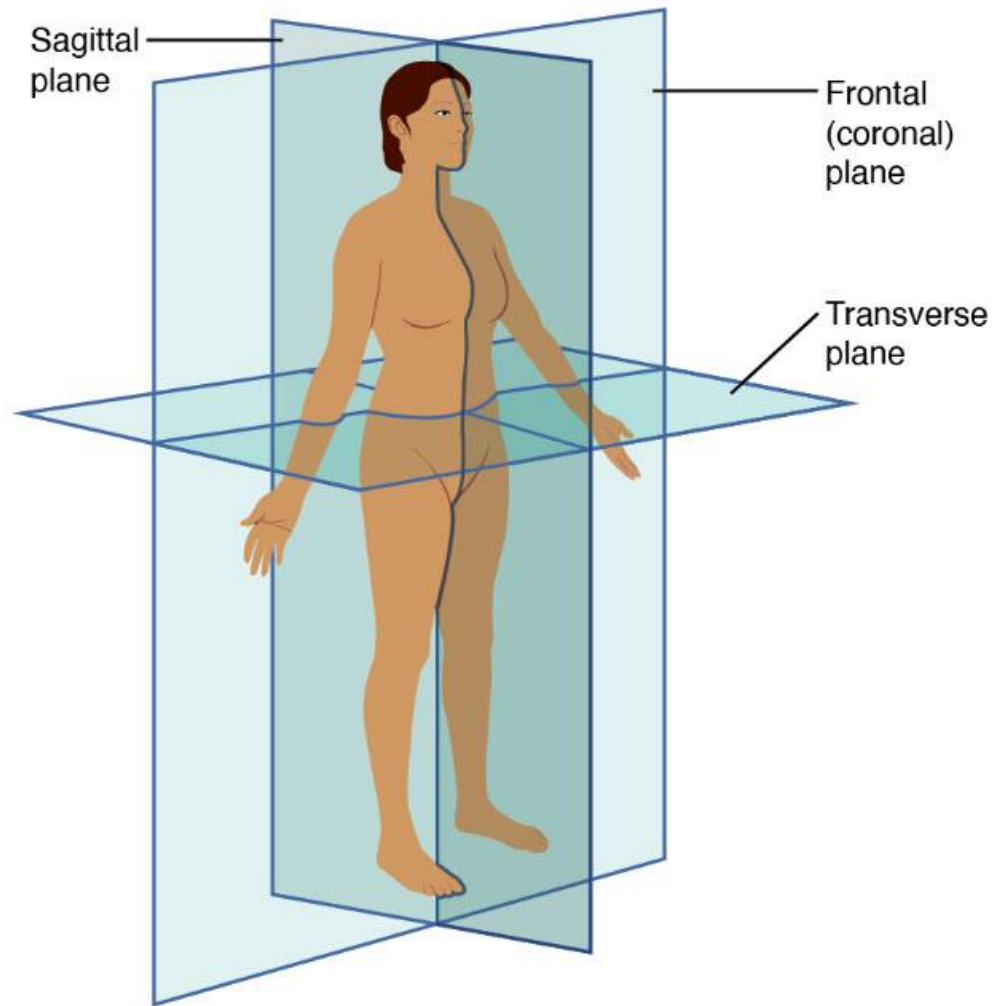


Fig. 1.15 Dorsal and Ventral Body Cavities The ventral cavity includes the thoracic and abdominopelvic cavities and their subdivisions. The dorsal cavity includes the cranial and spinal cavities.

➤ Planes of the Body

- **Coronal/Frontal plane**
 - Divides the body or organ vertically into anterior and posterior portions
 - Results in coronal section
- **Transverse plane**
 - Divides body or organ horizontally into superior and inferior portions
 - Results in a cross-section
- **Sagittal plane**
 - Divides the body vertically into left and right sides
 - Midsagittal plane: divides into equal left and right halves.

➤ Body Cavities

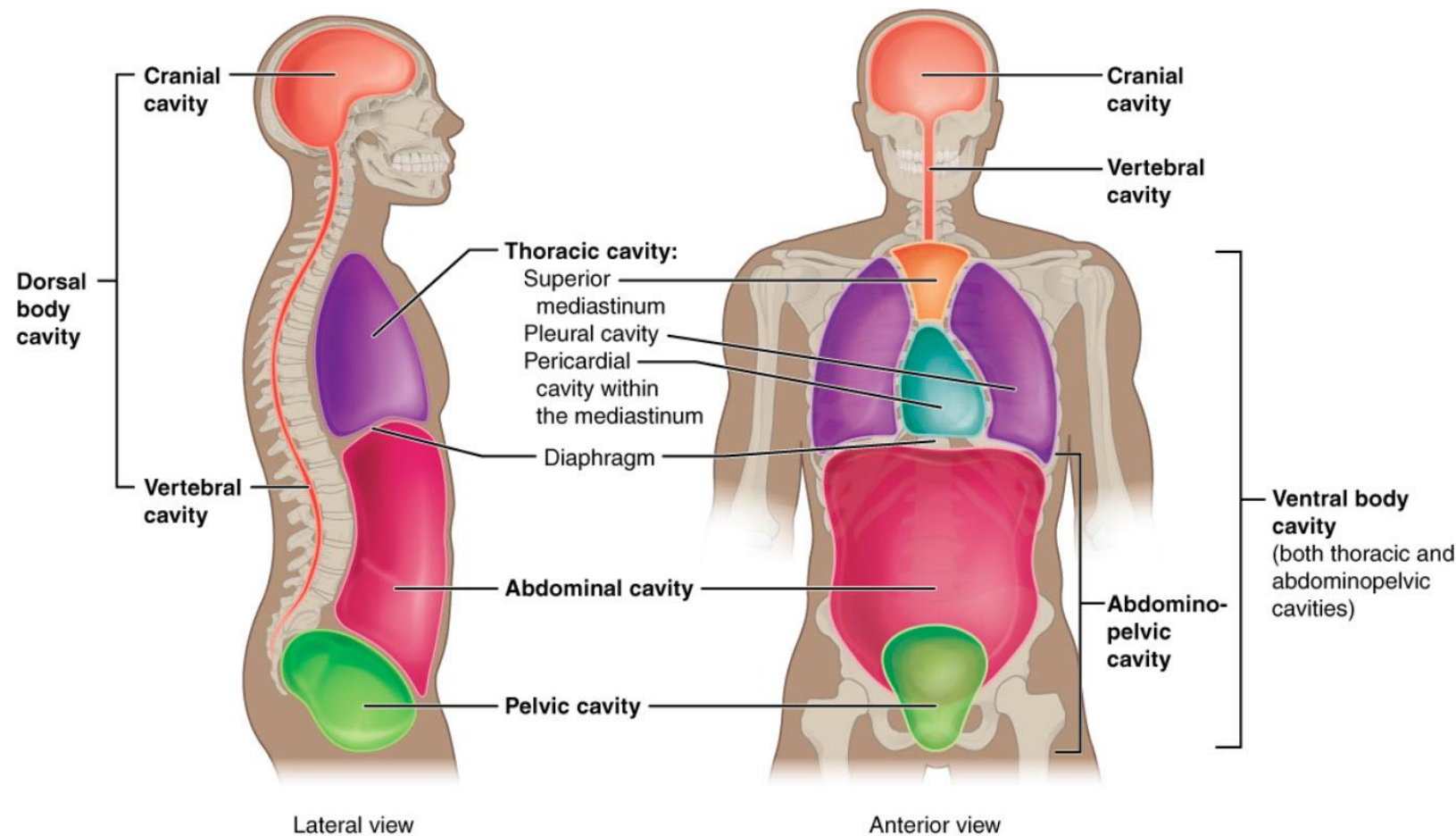


Fig. 1.15 Dorsal and Ventral Body Cavities The ventral cavity includes the thoracic and abdominopelvic cavities and their subdivisions. The dorsal cavity includes the cranial and spinal cavities.

➤ Abdominal Quadrants

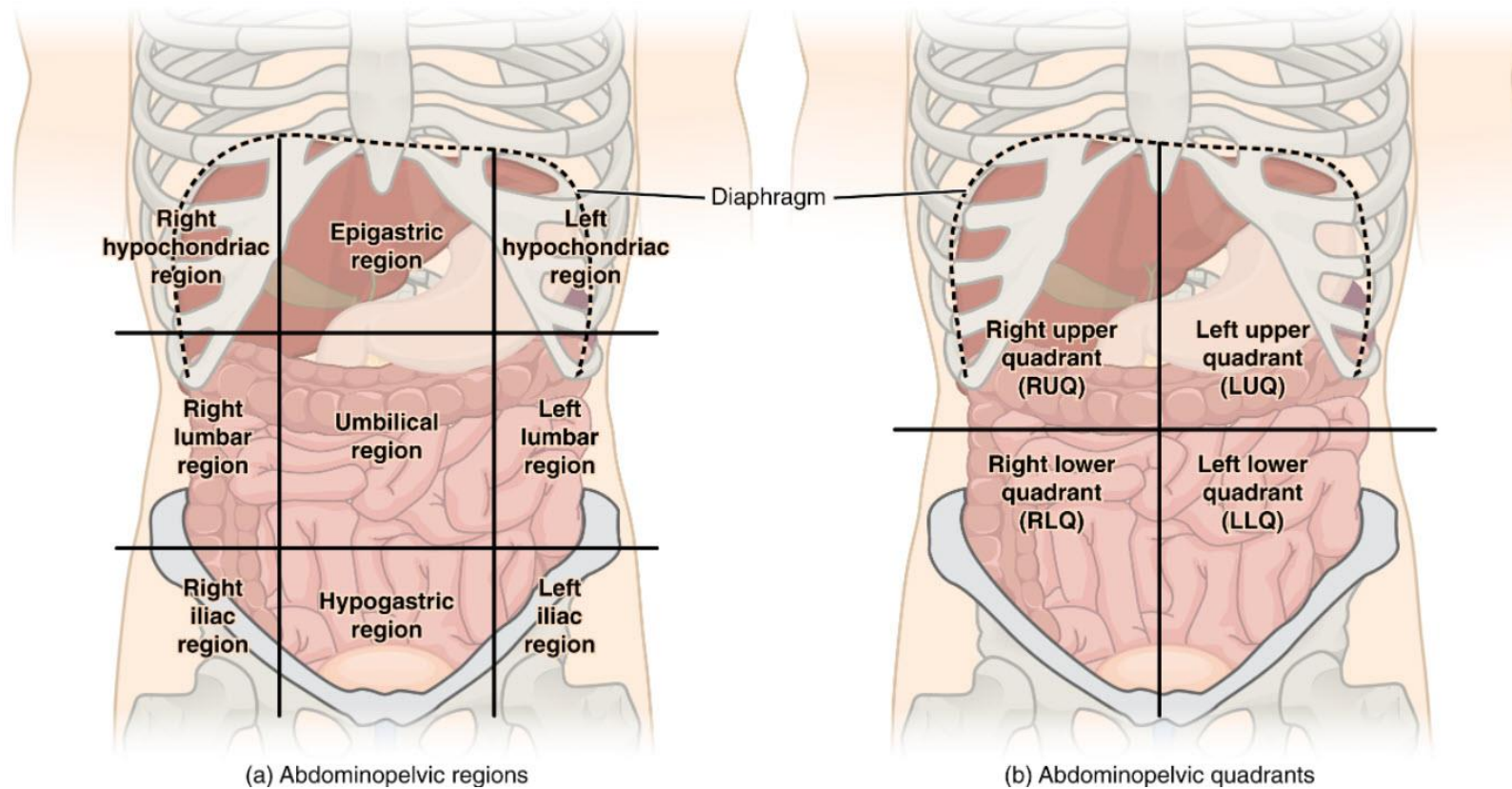


Figure 1.16 Regions and Quadrants of the Peritoneal Cavity There are (a) nine abdominal regions and (b) four abdominal quadrants in the peritoneal cavity.

➤ Types of Serous Membranes

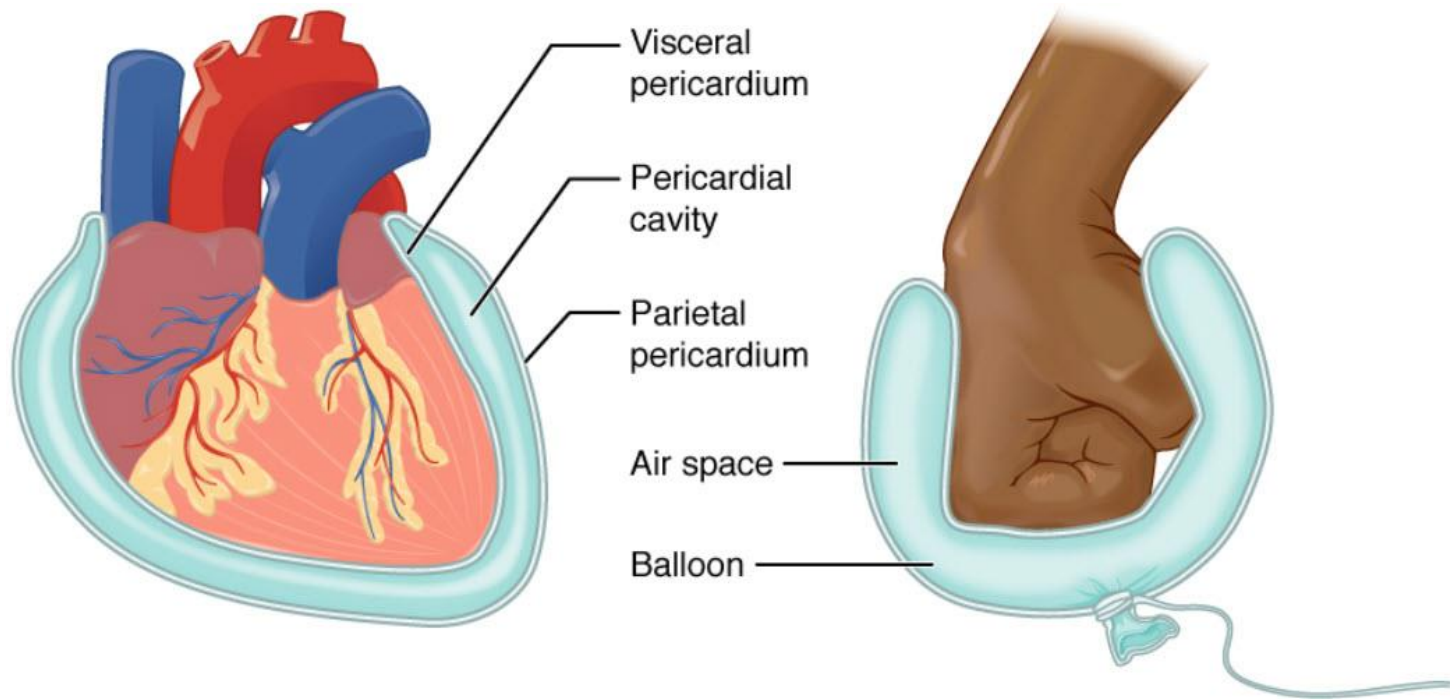


Fig. 1.17 Serous Membrane Serous membrane lines the pericardial cavity and reflects back to cover the heart—much the same way that an underinflated balloon would form two layers surrounding a fist.

➤ Summary

- Overview of the human body
- Homeostasis and negative feedback
- Anatomical and directional terms

➤ Next Lesson

- Chapter 2 (The Chemical Level of Organization)